Dynamic Loading Assessment at the Fifth Metatarsal in Elite Athletes With a History of Jones Fracture

Kenneth J. Hunt, MD,* Yannick Goeb, MD,† and Jonathan Bartolomei, MS*

Abstract

Objective: We hypothesize that athletes who have suffered Jones fractures will apply significantly higher loads at the fifth metatarsal base during athletic activities compared with matched uninjured athletes. **Design:** Sixteen athletes were recruited to participate. Eight athletes had a history of Jones fracture, and 8 age, gender, and position-matched athletes without a history of foot injury were recruited as controls. **Setting:** Institutional study at Stanford University. **Participants:** Sixteen athletes with/without a history of foot injury from Stanford University. **Interventions:** Athletes performed a standardized series of movements while wearing calibrated, wireless pressure mapping insoles, and then again with their custom corrective insoles. **Main Outcome Measures:** Peak pressure, mean pressure, maximum force, and force-time integral (ie, impulse) were recorded for each activity. **Results:** Athletes with a history of Jones fracture showed a significantly increased peak pressure (183 ± 23 vs 138 ± 7 kPA), mean pressure (124 ± 14 vs 95 ± 4 kPA), and maximum force (15 ± 1.2 vs 12 ± 1.2%BW) at the fifth metatarsal base during walking and running compared with uninjured matched controls (all $P < 0.05$). **Conclusions:** Athletes with a history of Jones fracture exert significantly increased peak and mean forces at the base of the fifth metatarsal during common athletic activities. Custom orthoses do not seem to offload this region in all cases. Increased loads may contribute to the development of stress injury to the fifth metatarsal during repetitive loading, and ultimately fracture of the bone.

Key Words: Jones fracture, dynamic loading, elite athletes, novel emed, orthotics, peak pressure, mean pressure, maximum force

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INTRODUCTION

Foot and ankle injuries are common in the athletic population, particularly in athletes participating in field and court sports. These injuries can often affect performance, result in time missed from participation, and in some cases require surgical intervention. Although data exist describing static risk factors for foot and ankle injuries,[1–3] little is known about dynamic mechanical risk factors for injuries to the foot and ankle during sports activities. Acute injuries cannot always be prevented, but there are several risk factors, associated with both acute and chronic foot and ankle injuries, which merit the attention of the team and treating physicians.[4] Identification of mechanical risk factors in athletes during sports activities can be a challenge, because most current loading assessment parameters are static (ie, radiographs, clinical exam, and pedobarographic mats). Advances in plantar pressure insole technology allow for the accurate quantification of dynamic loads at specific sites in the foot during athletic activities.[5]

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*From the *Department of Orthopedics, University of Colorado School of Medicine, Aurora, Colorado; and †Department of Orthopedics, Hospital for Special Surgery, New York, New York.*

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Corresponding Author: Kenneth J. Hunt, MD, Department of Orthopedic Surgery, University of Colorado School of Medicine, 1635 Aurora Ct, 4th Floor, Aurora, CO 80045 (Kenneth.J.Hunt@ucdenver.edu).

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Fractures of the proximal metaphysis of the fifth metatarsal (ie, Jones fractures) are a relatively common injury in athletes. They typically require surgical repair to reduce rates of recurrence and nonunion.[6] The underlying pathophysiology of the Jones fractures is in part related to repetitive loading of the bone with continuous physical activity, resulting in an incomplete remodeling process of the loaded bone which can ultimately fracture, even with relatively normal physiologic loads.[7–9] Subtle hindfoot varus alignment, which ostensibly results in larger forces at the fifth metatarsal, has been associated with increased risk of Jones fractures.[10] The concept that athletes with Jones fractures experience higher loads at the fifth metatarsal base with athletic activities has not yet been rigorously examined.

Our objective is to characterize the dynamic loads at the fifth metatarsal, during regular athletic activities, in elite athletes with a history of Jones fracture. We hypothesize that athletes who have suffered Jones fractures will demonstrate significantly higher loads at the fifth metatarsal base during sporting activities compared with matched athletes without a history of a foot injury. A better understanding of dynamic loading at the base of the fifth metatarsal may allow for the identification of athletes at risk for Jones fracture, aiding in the development of appropriate preventive devices and training regimens.

MATERIALS AND METHODS

Recruitment of Athletes

Sixteen elite, competitive athletes (12 football, 2 soccer, and 2 golf) were recruited for participation in this study. Eight of the 16 athletes (6 football, one soccer, and one golf) had been successfully treated with intramedullary screw fixation for

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Clinical Relevance

Identifying patients at risk for Jones fracture using dynamic loading assessment may facilitate an injury prevention program.

Jones fracture (Figure 1). All Jones fracture athletes had healed their fractures and recovered to a full level of function before participation in the study. The remaining 8 subjects were age, gender, and position-matched control athletes without any prior history of a foot injury and no other injuries within the past year. Controls were matched within 3 years of age of their corresponding Jones fracture athletes. Before participation, each athlete completed a validated clinical function survey [Foot and Ankle Ability Measure (FAAM) sport] to assess their current level of function.[11] Athletes used a standard athletic shoe during data collection. No cleats, spikes, or turf shoes were used by any athletes in the data collection process.

Data Collection

Clinical Variable

The FAAM sports subscale11 was administered to each athlete to ensure full function at the time of participation in the study. Means and SDs for each group were calculated.

Dynamic Loading Data Collection

While wearing calibrated, wireless pressure insoles (Novel Pedar X, 99 Hz St. Paul, Minnesota) in each shoe, all athletes performed a standardized series of 5 athletic movements. The movements performed included: (1) Treadmill walking, and (2) Treadmill running at pre-determined speeds (1.34 m/s for walking and 2.91 m/s for running), (3) Two-footed jumping with a required 2-footed takeoff and landing, (4) A 10-m shuttle run to simulate quick reversals of direction, and (5) A side-cut zig-zag course to mimic cutting at a 45-degree angle at speed (Figure 2). Movement tasks were selected based on typical athletic movements frequently experienced during training and competition. Athletes were instructed to perform the jump, shuttle, and cutting tasks at 90% of maximum effort. Each task was performed with a standard shoe, and again with the athlete's custom corrective orthotic insole (in Jones fracture athletes only).

Isolation of Fifth Metatarsal Base Data

Data from each completed task were analyzed using Novel Pedar software. Custom masks were created to isolate plantar loading data from the whole-foot pressure recording. For the purposes of our analysis, we isolated the loading forces and patterns at the base of the fifth metatarsal. Specific phases of each movement task were identified using live video recordings of athletes performing each movement. Novel Pedar software was used to view the plantar pressure recording as a series of steps (see **Figure 1**, **Supplemental Digital Content 1**, http://links.lww.com/JSM/A238). Steps identified in Pedar software were matched with steps seen on live video recordings to ensure each analyzed condition corresponded to the phases of interest for each movement. For running and walking tasks, the middle 10 steps of the trial were isolated for data analysis. This was done to ensure that the athletes had time to normalize their gait patterns after being fitted with the insole system. For the jumping task, the takeoff and landing phases were isolated. For shuttle and cutting tasks, the steps representing the change in direction ("cutting steps"), and the preceding and subsequent 2

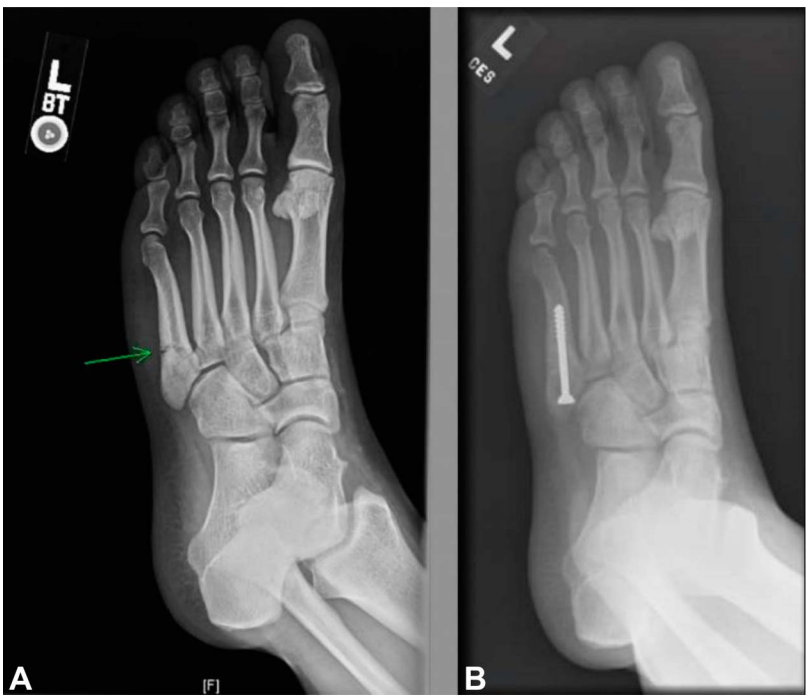

Figure 1. Representative radiograph of an athlete treated for Jones fracture (green arrow). A, Before surgical fixation and (B) 12 weeks postoperative.

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Figure 2. Images of an athlete wearing Novel Pedar system completing (A) shuttle cut task and (B) side cut task.

steps, were isolated. This allowed capture of the full initiation and completion of the change of direction.

Dynamic Loading Variables

After isolating each phase of interest from each movement task, values were recorded and means and standard deviations calculated for each of the following variables:

1. Peak pressure (kPA)
2. Mean pressure (kPA)
3. Maximum force (%BW)
4. Force–time integral (ie, impulse, %BW × second)

Data Analysis

Maximum force and force–time integral were normalized to body weight and expressed as %BW and %BW × second, respectively. Pair-wise comparisons between athletes with a history of Jones fracture and uninjured controls were performed for all variables using a paired Wilcoxon Sign Rank test using commercially available statistical analysis software (R, Vienna, Austria).

RESULTS

Clinical Comparison

Foot and Ankle Ability Measure Sport

All participating athletes reported a full level of function on the FAAM sport survey. The mean FAAM sport subscale score was 99.1 for athletes with a history of Jones fracture (97-100), and all controls scored 100. One Jones fracture athlete reported "slight difficulty with cutting/lateral movements" and another reported "slight difficulty performing sport for desired time." There was no statistically significant difference in FAAM sport subscale score between groups ($P = 0.17$).

Dynamic Loading at the Base of the Fifth Metatarsal

The highest magnitudes of body-weight adjusted mean and maximum pressures at the base of the fifth metatarsal were during cutting and running, followed by walking. Jumping and crossover (ie, shuttle run) showed much lower magnitudes in both groups. Of interest, the impulse was higher during walking than during running in both groups ($P = 0.04$ in Jones group and $P = 0.03$ in control group). Impulse was lowest for both groups during jumping and shuttle run (all $P < 0.05$ when jumping and shuttle impulses compared individually with walking, running, and cutting impulses).

Forces During Walking

There was a significantly higher peak pressure (112 ± 19 vs 89 ± 8 kPA), mean pressure (70 ± 12 vs 58 ± 4 kPA), and maximum force (8.67 ± 1.1 vs 7.42 ± 0.8 %BW) during walking in the Jones fracture group compared with controls ($P < 0.05$).

Forces During Running

Athletes with a history of Jones fracture showed a significantly higher peak pressure (183 ± 23 vs 138 ± 7 kPA), mean pressure (124 ± 14 vs 95 ± 4 kPA), and maximum force (15 ± 1.2 vs 12 ± 1.2 %BW) at the fifth metatarsal base during running compared with uninjured matched controls (all $P < 0.05$). A 30% increase in force–time integral (impulse) was observed at the fifth metatarsal in Jones fracture athletes during running, but this did not reach a statistically significant difference between the groups ($P = 0.07$).

Forces During Cutting

We found a significant increase in maximum force and force–time integral in Jones fracture athletes during cutting ($P < 0.05$). There was a 21% higher peak pressure and a 27% higher mean pressure at the fifth metatarsal base in Jones fracture athletes compared with uninjured controls during cutting, but this did not result in a statistically significant difference (both $P = 0.07$).

Forces During Jumping

There was no statistically significant difference in mean pressure, peak pressure, maximum force, or impulse at the fifth metatarsal base during jumping between athletes with and without a history of Jones fracture (Figure 3).

Effect of Custom Insoles

In Jones fracture athletes, use of a custom, full length, semi-rigid corrective orthotic insole did not significantly reduce peak or mean pressure, maximum force, or impulse at the fifth metatarsal base during walking, running, jumping, or cutting (Figure 4).

DISCUSSION

Fractures of the fifth metatarsal proximal metaphysis, that is, Jones fracture, are a relatively common athletic injury, and

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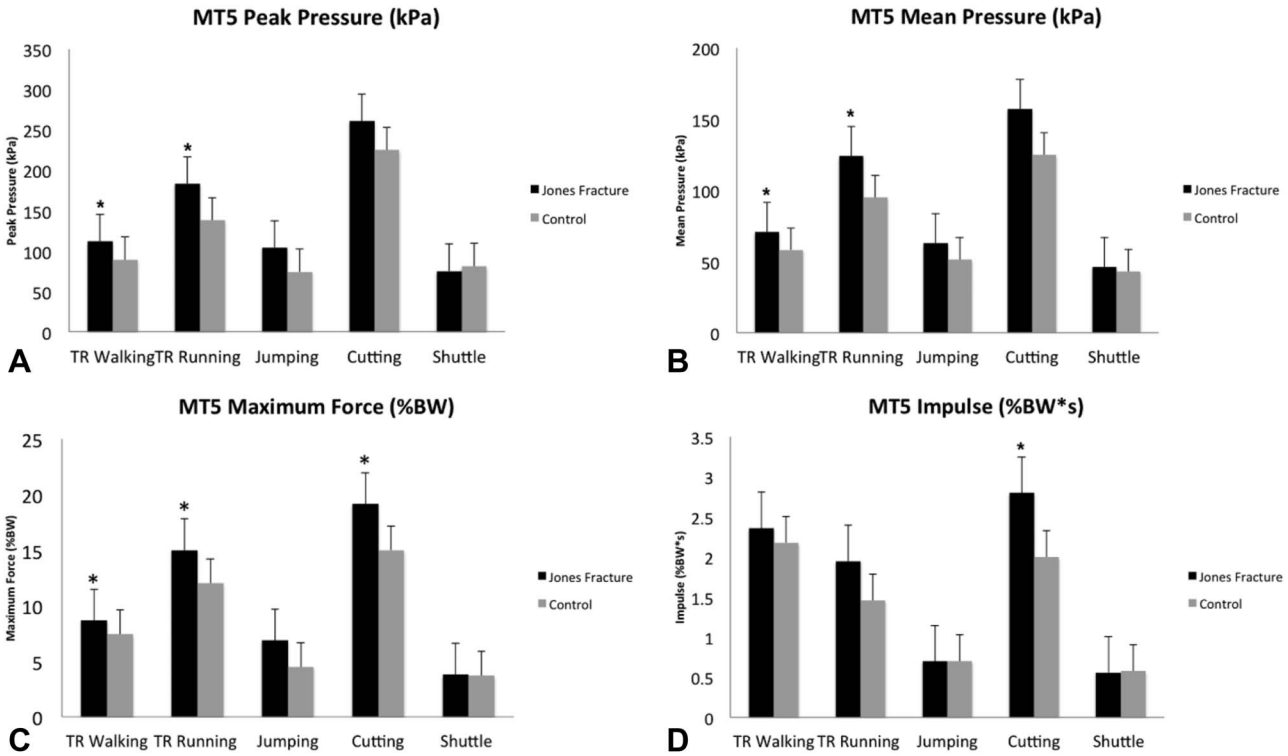

Figure 3. Summary data showing (A) peak pressure (B) mean pressure (C) maximum force and (D) impulse at the fifth metatarsal during 5 common athletic movements. *Indicates a significant difference at the 5% level between athletes with a history of Jones fracture (black bars) and uninjured controls (grey bars).

many athletes with Jones fractures show clinical and radiographic evidence of stress fractures. Our data demonstrate that athletes with a history of Jones fracture apply higher loads to the base of the fifth metatarsal during walking, running, and cutting relative to uninjured controls. It is plausible that repetitive exposure to these increased loads leads to an increased risk in fifth metatarsal bone stress injury. It is important to note that a significant difference in loads at the fifth metatarsal between Jones fracture athletes and uninjured controls was found in both walking and running, the most common movements performed in field and court sports, training, competition, and daily activities. The highest pressures and forces were seen during cutting, and Jones fracture athletes showed significantly increased maximum force and impulse at the fifth metatarsal in this movement task. The combination of increased loads at the fifth metatarsal during repetitive, common movements, such as running, and increased forces at the fifth metatarsal during more explosive movements, such as cutting, may have contributed to the development of Jones fracture.

Our data do not suggest a significant difference in forces at the fifth metatarsal between groups during the shuttle activity. The overall relatively small loads observed at the fifth metatarsal during the shuttle activity may be explained by the deceleration necessary before changing direction with a shuttle cut. At the time of the shuttle cut, when data were

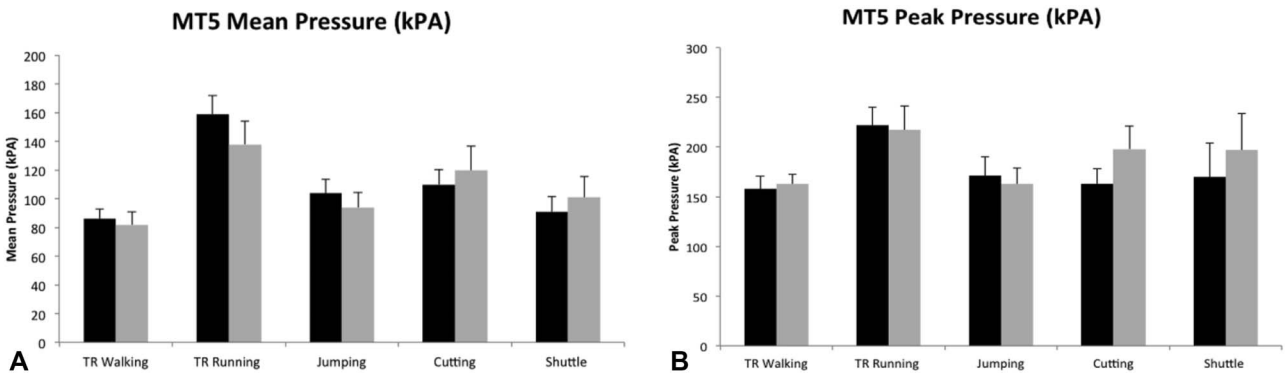

Figure 4. Representative data showing (A) peak pressure and (B) mean pressure at the fifth metatarsal during 5 common athletic movements. Black bars represent patients with a history of Jones fracture without corrective orthotics. Grey bars represent the same athletes tested while wearing corrective orthotics.

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sampled, the athletes had reduced their running speed considerably, and preferentially loaded the medial aspect of their planted cutting foot to achieve maximal push-off during the change of direction. Consistent with our data, a recent study investigating the loading patterns of soccer players during various athletic movements found that side cutting (similar to our "shuttle" activity) places increased loads on the medial rather than lateral forefoot.[12]

The magnitude of forces demonstrated in the present study is far below the forces required to fracture a healthy fifth metatarsal bone. Biomechanical studies[13] of the fifth metatarsal suggest that a healthy bone is able to tolerate a maximum pressure to mechanical failure of approximately 1100 kPA, a value much larger than those observed in our data and other studies examining fifth metatarsal pressure in athletic activity.[14–16] Yet, many Jones fractures occur during normal sports activities without substantially supraphysiologic loads. We hypothesize that prolonged periods of repetitive bone loading at higher magnitudes, as demonstrated in our Jones fracture group, accumulate stress and sufficiently weaken the bone resulting in fracture at much smaller forces and stresses than would be necessary for a complete acute fracture.

In addition to the magnitude of the force, the location of force and impulse are also likely contributors to fracture risk. Our findings that the loads at the base of the fifth metatarsal are higher in running and cutting (and walking, to some extent) than they are during jumping supports the higher incidence in sports with frequent running and cutting (ie, football and basketball) and lower in predominantly jumping sports (ie, volleyball). We also found that impulse is much lower in jumping and the shuttle cut activities than the other movement tasks studied. It is likely the combination of higher mean and maximum loading forces, combined with higher impulse during running, cutting, and walking, that contribute to risk. The Jones fracture occurs at the diaphyseal–metaphyseal junction, which is not the greatest point of impact, but is where tension stresses accumulate (with contributing forces from the peroneus brevis) potentially leading to a fracture.

Another important finding of our study is that custom corrective orthotic devices failed to reduce loading at the fifth metatarsal in every testing condition studied. It is unclear whether this is due to the design of the orthoses, the material used, or whether corrective insoles do not inherently alter gait patterns to reduce loads at the lateral foot effectively. Every athlete treated for Jones fracture at our institution is provided with custom corrective orthoses designed for use during training and competition. Our results suggest that these orthoses may not sufficiently offload the lateral foot and may do little to prevent increased loads at the site of the previous fracture. A modification in insole design to more effectively elevate the lateral column of the hindfoot was available for 2 athletes in this study. These athletes had a 15% reduction in peak pressure at the fifth metatarsal during walking and running, but not cutting or jumping. This observed difference was not statistically significant. Additional study of a newer generation of orthoses in this population is necessary to determine the clinical and biomechanical effectiveness of custom insoles built to offload the base of the fifth metatarsal.

The most significant limitation of our study is the fact that Jones fracture athletes had already experienced the injury before data collection. It is therefore impossible to determine whether any observed differences between Jones fracture athletes and uninjured controls are due to true differences in the loading patterns of athletes with increased risk of Jones fracture or if the observed differences are reactive mechanisms developed by the Jones fracture athletes postinjury. It seems unlikely that athletes with a previous Jones fracture would develop an altered gait pattern that increases loads at the fifth metatarsal postinjury, but our experimental design prevents us from entirely discounting this possibility. In addition, our study only collected data related to the foot. Any biomechanical differences proximal to the foot were not recorded and may have contributed to compensatory movements, both in our injured and noninjured athlete groups, which may have altered the outcome of the study. A long-term, prospective study is underway at our institution examining the dynamic loading distributions of healthy, uninjured athletes as a risk factor for Jones fractures and other chronic and acute injuries.

A second limitation of our study is the variability associated with allowing athletes to select their own training shoe to wear during the data collection phase. Studies have shown differences in plantar pressure distribution patterns in different types of footwear,[17] and although all athletes in our study wore a similar brand, type, and style of shoe (ie, training shoe as opposed to cleats), the slight differences in footwear design and wear may result in altered plantar pressure loading patterns. The decision to allow athletes to wear a training shoe of their choice, largely to ensure comfort and familiarity with the footwear worn during data collection, was made to ensure true movement patterns. Athletes given a new shoe to wear often complain about improper fit and insufficient time to "break in" the shoe for maximal comfort. A previously published case study of 23 Jones fractures showed that 35% of the fractures were sustained in cleated sports.[18] Most (12/16) athletes tested in this study were football players, who often train in cleats, but were tested in training shoes. Further studies are needed to determine whether a significant difference exists between loading patterns in cleats and standard training shoes in elite athletes.

The goal of our work is to identify potentially modifiable risk factors for Jones fracture risk in elite athletes. Previous studies show that recent changes in footwear or training surface are risk factors for the development of stress fractures.[19,20] Together with these risk factors, subtle hindfoot varus resulting in increased lateral forefoot loading may further exacerbate stress fracture risk. We are in the process of following athletes, having undergone correction of mechanical risk factors with new generation insoles, to determine whether there will be a reduction in fracture incidence.

CONCLUSIONS

Our work demonstrates significant differences in loading patterns at the fifth metatarsal between athletes with a history of Jones fracture and uninjured controls. Future studies examining the connection between dynamic loading distribution and foot and ankle injury risk in athletes should attempt to put the increased loads seen in specific movement tasks in a biomechanical context. The increased loads seen in any particular region of the foot are likely due to subtle foot alignment differences, which may manifest as increased loads in certain areas and decreased loads in others. Attempting to associate these patterns of altered loading along the plantar surface with foot and ankle injury risk requires advanced statistical analysis that can readily analyze high dimensional data sets.

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